

Introduction to Algorithms (Interactive Edition)

Author: Thomas H. Cormen | Category: Computer Science

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Part I: Foundations of Algorithms

Algorithm analysis: Big-O, Omega, and Theta notation definitions.

Divide-and-conquer recurrences solved using the Master Theorem.

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Part II: Sorting and Order Statistics

Heapsort, Quicksort, and linear time sorting algorithms.

Lower bounds for comparison sorting: why $O(n \log n)$ is optimal.

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Part III: Advanced Data Structures

Red-Black Trees, B-Trees, and Fibonacci Heaps.

Amortized analysis techniques: aggregate, accounting, and potential methods.

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Part IV: Graph Algorithms

Breadth-First Search (BFS) and Depth-First Search (DFS) traversals.

Minimum Spanning Trees: Kruskal and Prim algorithms with priority queues.

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Part V: NP-Completeness

Polynomial time reductions and proof techniques.

Classic NP-complete problems: SAT, Vertex-Cover, Traveling Salesperson.